REPORT

Of

C++ Programming Internship

Object-Oriented and File-Handling Based Console Applications using C++

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

Bachelor of Technology (B.Tech CSE)

By

Name of Student	AAKHYAT ANAND
Roll Number	25SCS1003004645
Batch	2025-2029
Internship Organisation	Thiranex EduTech LLP — Skill Development & Future Tech
Internship Duration	30 August 2026 – 29 September 2026

2026-2027

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1. CANDIDATE'S DECLARATION

I, **Aakhyat Anand**, do hereby declare that the internship report titled **"C++ Programming Internship — Object-Oriented and File-Handling Based Console Applications using C++"** has been completed by me to fulfil the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**. The report presents the work carried out and the knowledge acquired during my internship as a **C++ Programming Intern at Thiranex EduTech LLP**, from 30 August 2026 to 29 September 2026. The internship involved the development of four C++ projects — a Student Management System, a Bank Management Application, a Library Management System, and a Mini Game Project (Tic-Tac-Toe / Snake) — and provided practical experience in object-oriented programming, file handling, menu-driven program design, data structures, and debugging and testing.

All references and sources used in the preparation of this report have been appropriately acknowledged. I affirm that this report represents my own work and has not been submitted to any other institute for any degree or diploma requirement. I shall be solely responsible for any kind of copyright violation in this regard.

Signature:

Aakhyat Anand

Student Name: Aakhyat Anand
Roll No.: 25SCS1003004645
Programme: B.Tech Computer Science and Engineering (Batch 2025–2029)

2. ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout my internship programme. This internship gave me valuable practical exposure to **C++ programming** and helped me understand how the core concepts of the language — classes and objects, file handling, control flow, and data structures — come together to build complete, working applications. I am thankful to **Thiranex EduTech LLP** for giving me the opportunity to work as a **C++ Programming Intern** and to learn through real, project-based work.

I am grateful for the guidance and industry mentorship provided during the internship, together with the periodic progress reviews, which helped me improve my understanding of **object-oriented design, persistent storage using file handling, menu-driven program flow, and systematic debugging**. Building the Student Management System, Bank Management Application, Library Management System, and Mini Game Project helped me understand the full life cycle of a software project, from understanding requirements to design, implementation, testing, and submission of the source code.

I would also like to express my gratitude to the faculty members of the **School of Computer Science and Engineering, IILM University, Greater Noida**, for their academic guidance and support in preparing this internship report.

Finally, I express my heartfelt gratitude to my **parents** and **friends** for their constant encouragement, support, and blessings throughout my academic journey and internship. Their motivation has always inspired me to learn new skills, overcome challenges, and give my best in every opportunity.

Signature:



Student Name: Aakhyat Anand
Roll No.: 25SCS1003004645
Programme: B.Tech Computer Science and Engineering (Batch 2025–2029)

3. INTERNSHIP COMPLETION CERTIFICATE



Figure 1: Internship Completion Certificate issued by Thiranex (Verified Certificate ID: THX-AUG3126-002)

4. PROJECT DESCRIPTION

4.1 Introduction

The C++ Programming Internship at Thiranex is a remote, project-based programme designed to give students practical experience in building real, working software applications. Rather than limiting learning to theory, the internship is organised around four hands-on projects, each of which applies a different set of C++ concepts: a Student Management System, a Bank Management Application, a Library Management System, and a Mini Game Project (Tic-Tac-Toe / Snake).

The main idea behind the internship is to help an intern move from understanding individual C++ features to combining them into complete, usable applications. Each project builds on the one before it: the first establishes record management with file handling and menu-driven navigation; the second introduces object-oriented design in a banking context; the third extends this to multiple related entities with issue, return and search operations; and the fourth applies loops, arrays and conditional logic to build an interactive game.

The first three projects focus on data management, in which user records are created, modified, searched and stored persistently in files. The Mini Game Project shifts the focus to program logic and control flow, using a dynamically redrawn board, win/loss detection and replay support. Together, the projects provide practical exposure to object-oriented programming, file handling, menu-driven design, data structures, and debugging and testing.

The overall aim of the internship is to build a strong programming foundation in C++ that can later be extended to larger, more advanced work such as database-backed applications, graphical interfaces and software engineering practice.

4.2 Organization Profile

The internship was completed under the guidance and supervision of Thiranex EduTech LLP. The key details of the internship are summarised below.

Organization Name	Thiranex EduTech LLP — Skill Development & Future Tech
Internship Role	Intern – C++ Programming
Internship Duration	30 August 2026 to 29 September 2026 (4 weeks)
Work Mode	Remote / Project-Based
Intern ID	THX-AUG3126-002
Offer Letter Date	30 August 2026
Certificate Signed By	Hariharan M, Founder & CEO

Table 1: Internship details

Thiranex EduTech LLP is a skill development and future technology organization registered as an MSME (Ministry of Micro, Small & Medium Enterprises) under **UDYAM-TN-02-0512885**, with its office at No. 62, Balakrishnan Street, Vyasarpadi, Chennai, Tamil Nadu 600039. The organization provides a practical learning environment in which interns work on defined

projects under industry mentorship. Progress is reviewed periodically, and an intern's performance determines the successful completion of the internship.

During the internship, the work focused on understanding the requirements of each project, designing its structure, implementing it in C++, testing its behaviour with different inputs, and submitting the source code for review. The internship provided exposure to how small software applications are planned, built, and refined.

4.3 Problem Statement

Many engineering students learn the syntax of C++ as part of their coursework, but get little structured practice in turning that knowledge into complete applications. Concepts such as classes, encapsulation, file handling and data structures often remain abstract until they are applied to realistic problems.

A second challenge is data persistence. Information held only in variables is lost as soon as a program ends, whereas practical applications such as student databases, banking systems and library catalogues must retain their records between runs. Handling this correctly requires an understanding of file input and output, and of reading and writing structured records reliably.

Third, applications must be usable. A program with many features is of little value if its users cannot navigate it. Designing clear, menu-driven interfaces, validating user input, and handling invalid or unexpected input gracefully require deliberate practice.

Finally, programming logic must be correct. Features such as issuing and returning books, validating a withdrawal against an available balance, or detecting a winning move in a game all depend on careful reasoning about conditions and edge cases, which is developed mainly through debugging and testing.

This internship addresses these problems by requiring the development of four practical C++ projects that progressively combine object-oriented design, persistent storage, menu-driven interaction, and game logic. The main objective is therefore to turn theoretical knowledge of C++ into practical, working applications.

4.4 Project Objectives

The primary objective of the internship is to develop practical C++ programming skills by building complete, working console applications. The major objectives are:

1. C++ Programming Fundamentals

To apply variables, control structures, functions, arrays and strings to solve practical problems.

2. Object-Oriented Programming

To model real-world entities such as accounts, books and members using classes, objects and encapsulation.

3. File Handling and Persistent Storage

To store and retrieve records using files so that application data survives between program runs.

4. Menu-Driven Program Design

To build clear, navigable console interfaces through which users can perform operations easily.

5. Data Structures and Record Management

To organise related data using structures, classes and arrays, and to support operations such as add, update, delete, search and display.

6. Game Logic Development

To implement dynamic board display, win/loss detection and replay options using loops, arrays and conditional statements.

7. Debugging and Testing

To test each application systematically, identify defects, and correct them using the feedback received during periodic reviews.

8. Code Organization and Documentation

To maintain well-structured, readable code and to document the completed work in a presentation and this report.

4.5 Scope of the Project

The scope of the internship covers the design and development of four C++ applications, each focusing on a different set of programming concepts. The projects were assigned with weekly deadlines, and the complete source code of all four projects is maintained in a project repository.

No.	Project	Focus Area	Due Date
1	Student Management System	File handling, record management, menu-driven operations	06 Sept 2026
2	Bank Management Application	Object-oriented programming, transactions, file management	13 Sept 2026
3	Library Management System	Classes/structures, issue and return, search by title or author	20 Sept 2026
4	Mini Game Project (Tic-Tac-Toe / Snake)	Loops, arrays, conditional logic, win/loss detection	27 Sept 2026

Table 2: Summary of internship projects

Project 1: Student Management System

Objective: A console-based Student Management System in C++ that efficiently manages student records using file handling and menu-driven operations.

Key features:

- Add, update, delete, and display student information.
- Use file handling to store data persistently, so that records remain available between program runs.
- Implement menu-driven operations for smooth navigation.

Concepts applied: Structures/classes for student records, file input/output streams, loops and switch-based menus, and basic input validation.

Expected outcome: A functional application that allows users to add, update, delete, and display student information with persistent data storage for reliable record management.

Due date: 06 September 2026

Project 2: Bank Management Application

Objective: A C++ based Bank Management Application that simulates core banking operations using object-oriented programming and file handling.

Key features:

- Include features like deposit, withdrawal, and balance check.
- Use object-oriented programming concepts such as classes, encapsulation, and member functions.
- Store customer data securely using file management.

Concepts applied: Classes and objects, encapsulation of account data, member functions for transactions, file handling for customer records, and validation of transaction amounts.

Expected outcome: A secure and functional banking system capable of performing deposits, withdrawals, and balance inquiries while maintaining persistent customer records.

Due date: 13 September 2026

Project 3: Library Management System

Objective: A Library Management System that efficiently manages books, members, and borrowing records using structured or object-oriented programming.

Key features:

- Implement structures or classes for book and member details.
- Support book issue and return features.
- Include search functionality by title or author.

Concepts applied: Multiple related classes or structures, collections of records, searching techniques, issue/return state tracking, and persistent storage.

Expected outcome: A functional application that enables book addition, issue and return processing, and search by title or author for streamlined library operations.

Due date: 20 September 2026

Project 4: Mini Game Project (Tic-Tac-Toe / Snake)

Objective: A console-based mini game in C++ (Tic-Tac-Toe / Snake) that demonstrates core programming concepts such as loops, arrays, and conditional logic.

Key features:

- Use loops, arrays, and condition statements for game logic.
- Display the game board dynamically after each move.
- Add win/loss detection and replay options.

Concepts applied: Arrays for the game board, nested loops for rendering, conditional logic for win/loss detection, a game loop, and replay handling.

Expected outcome: An interactive game with dynamic board display, win/loss detection, and replay functionality, showcasing effective implementation of game logic.

Due date: 27 September 2026

Future Scope

The projects can be extended in future in several directions: replacing flat-file storage with a database such as MySQL or SQLite; adding user authentication, encrypted records and transaction history to the banking application; adding fine, due-date and reservation features to the library system; adding a score board, difficulty levels and a graphical interface to the game; building a graphical front end with a framework such as Qt; and adding automated unit tests to improve reliability.

4.6 Technologies and Tools Used

The internship projects were developed using standard C++ and its standard library, without the need for external frameworks.

Technology	Purpose and Usage	Used In
C++ Programming	Core language used to implement the application logic of every project.	All projects
Object-Oriented Design	Classes, objects, encapsulation and member functions used to model accounts, books, members and students.	Projects 1, 2, 3
File Handling	File input/output streams used to store and retrieve records so that data persists between runs.	Projects 1, 2, 3
Menu-Driven Logic	Loop and switch-based menus that guide the user through each available operation.	All projects
Data Structures	Structures, arrays and collections used to organise and search records and to represent the game board.	All projects
Debugging & Testing	Systematic testing with valid, invalid and boundary inputs, and correction of the defects found.	All projects
Standard Library	Input/output streams, strings and containers from the C++ Standard Library.	All projects

Table 3: Technologies and tools

A standard C++ compiler, a code editor or IDE, and a console/terminal were used to write, compile, run and debug the applications.

4.7 System Architecture

Each application follows a simple, layered console design in which responsibilities are divided between the user interface, application logic, data model, and storage. The layers build on one another from user input to persistent records.

No.	Layer	Responsibility
1	Interface Layer	Menu-driven console interface that displays options, accepts input and shows results.
2	Control Layer	Loops and switch statements that read the user's choice and call the relevant operation.
3	Logic Layer	Functions and class member functions implementing operations such as deposit, issue, search and win detection.
4	Data Model Layer	Structures and classes representing students, accounts, books, members and the game board.
5	Storage Layer	File handling routines that read records from and write records to files.

Table 4: Layers of the application architecture

4.7.1 Interface and Control Layers

Every application presents a main menu, reads the user's selection and dispatches to the corresponding function until the user chooses to exit. This keeps the program flow simple and predictable, and allows new features to be added as additional menu options.

4.7.2 Logic and Data Model Layers

Real-world entities are represented as structures or classes whose data members hold their details and whose member functions perform operations on them. In the banking application, data hiding keeps the account balance protected so that it can only be changed through the deposit and withdrawal operations. In the library system, separate structures or classes for books and members are linked through issue and return records.

4.7.3 Storage Layer

For the record-based applications, data is written to files whenever it changes and read back when the program starts or when a record is requested. This ensures that student, customer and library records remain available after the program is closed and reopened.

4.7.4 Game Architecture

The Mini Game Project follows the same principle with a game loop: the board is stored in an array, redrawn after each move, checked for a win, loss or draw, and then the next turn is processed. At the end of a game, the player is offered the option to replay.

Overall Data Flow

The general flow of the record-based applications can be represented as:

User → Menu Interface → Program Logic (Functions / Classes) → File Storage → Processed Data → Displayed Result

Keeping these responsibilities separate makes the applications easier to understand, debug, and extend.

4.8 Methodology

The internship followed a progressive, project-by-project approach. Each project introduced new concepts on top of those already learned, moving from understanding the requirements to designing, implementing, testing and finalising the application.

4.8.1 Requirement Analysis

For every project, the first step was to study the project brief and identify the required features. For example, the Student Management System required adding, updating, deleting and displaying records with persistent storage, whereas the Library Management System required book and member management, issue and return processing, and search by title or author.

4.8.2 Planning and Design

Before coding, the structure of each project was planned. This included deciding which structures or classes were needed, which operations each would support, how the menu would be organised, and which data had to be stored in files.

4.8.3 Implementation

Each application was implemented in C++ in stages: first the data structures or classes, then the core operations, then file handling, and finally the menu that ties the features together. Building the application incrementally made it easier to test each feature before moving on.

4.8.4 Object-Oriented Development

In the Bank Management Application and Library Management System, object-oriented principles were applied. Account, book and member details were encapsulated in classes, and operations such as deposit, withdrawal, issue and return were implemented as member functions.

4.8.5 File Handling

Files were used in the record-based projects to save data whenever it changed and to load it again when needed, so that records persisted between runs of the program.

4.8.6 Game Logic Development

In the Mini Game Project, loops and arrays were used to represent and display the board, conditional statements were used to check for a win, loss or draw after each move, and a replay option was added at the end of each game.

4.8.7 Testing and Debugging

After implementation, each application was run and tested to confirm that its features worked correctly. Testing focused on:

- Correct behaviour of each menu option and smooth navigation between them.
- Adding, updating, deleting and displaying records, and persistence of the records after restarting the program.

- Deposit, withdrawal and balance operations, including attempts to withdraw more than the available balance.
- Book issue, return and search operations, including books that are unavailable or not found.
- Correct display of the game board and correct win, loss and draw detection.
- Handling of empty, invalid or out-of-range input.

Errors found during testing were debugged, and the corresponding code was corrected. Feedback received during the periodic progress reviews was used to improve the logic and handle edge cases.

4.8.8 Review and Documentation

The final stage included reviewing the code, organising the project files, and summarising the work in an internship presentation and in this report. Progress was reviewed periodically under mentorship, and the source code of all four projects was submitted in a repository.

Skills and Concepts Applied

- **C++ Programming:** Functions, control flow, arrays, strings and the Standard Library.
- **Object-Oriented Design:** Classes, objects, encapsulation and member functions.
- **File Handling:** Persistent storage and retrieval of records using file streams.
- **Menu-Driven Logic:** Clear, navigable console interfaces for every application.
- **Data Structures:** Structures, arrays and collections to organise and search data.
- **Debugging & Testing:** Systematic testing and correction of defects, supported by mentor feedback.

4.9 Expected Outcomes

The expected outcome of the internship is a portfolio of four working C++ applications and a strong practical understanding of the concepts used to build them.

Applied Object-Oriented Programming

The ability to translate classes, encapsulation and structures into working systems for banking and library data.

Persistent Data with File I/O

Reliable storage and retrieval of records between runs in every record-based project.

Menu-Driven Program Design

Clear, navigable console interfaces so that each application is usable from start to finish.

Record Management Applications

Complete add, update, delete, display, issue, return and search workflows, as shown by the Student, Bank and Library projects.

Game Logic Implementation

Dynamic board display, win/loss detection and replay functionality, as shown by the Mini Game Project.

Problem-Solving Under Review

Iterating on logic and edge cases using mentor feedback across periodic progress reviews.

Overall, the internship resulted in practical C++ programming experience and a portfolio of applications that demonstrate the ability to design, build and test working software. It provides a solid base that can be extended to larger projects involving databases, graphical interfaces and advanced software engineering practice.

4.10 Certificates of Completion and Other Communication Proof

The internship completion certificate issued by **Thiranex EduTech LLP** confirms the successful completion of the internship in **C++ Programming** by **Aakhyat Anand**. The internship was completed from **30 August 2026 to 29 September 2026**. The certificate (Verified Certificate ID: **THX-AUG3126-002**) was signed by Hariharan M, Founder & CEO. During the internship, the work focused on four C++ projects: the Student Management System, Bank Management Application, Library Management System, and Mini Game Project (Tic-Tac-Toe / Snake).



Figure 2: Internship Completion Certificate issued by Thiranex

4.10.2 Internship Communication Proof



Thiranex
SKILL DEVELOPMENT & FUTURE TECH

Thiranex EduTech LLP
MSME Registration: UDYAM-TN-02-0512885
No. 62, Balakrishnan Street, Vyasarpadi, near Saraswathi Stores, Chennai, Tamil Nadu 600039

Date: 30 August 2026

TO,
Aakhyat Anand
Intern ID: THX-AUG3126-002

Subject: Offer Letter for Internship in C++ Programming

Dear **Aakhyat Anand,**

Following our selection process, we are pleased to offer you an internship position at **Thiranex EduTech LLP**. We believe your skills and enthusiasm will be a great addition to our team.

Internship Role	**Intern - C++ Programming**
Commencement Date	**30 Aug 2026**
Completion Date	**29 Sept 2026**
Work Mode	**Remote / Project-Based**

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.

Thiranex

This is an electronically generated document. No physical signature is required.

Thiranex EduTech LLP | UDYAM-TN-02-0512885 | www.thiranex.in

Figure 3: Offer Letter issued by Thiranex (Internship Communication Proof)

The offer letter dated 30 August 2026 was issued by Thiranex EduTech LLP to confirm the internship position of Intern – C++ Programming (Intern ID: THX-AUG3126-002), with a commencement date of 30 Aug 2026 and a completion date of 29 Sept 2026, on a remote, project-based work mode.

5. BIBLIOGRAPHY

The following resources were used as references during the development and documentation of the internship projects:

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4. cppreference.com — C++ Reference, including Input/Output Library (fstream) and Containers (en.cppreference.com).
5. Standard C++ Foundation — C++ Core Guidelines (isocpp.github.io/CppCoreGuidelines).
6. LearnCpp.com — Free C++ Tutorial Site (learncpp.com).
7. Thiranex EduTech LLP — C++ Programming Internship Programme Materials and Project Briefs.
8. Thiranex Internship Presentation — C++ Programming Internship Portfolio, Aakhyat Anand (Intern ID: THX-AUG3126-002).